

Linear Data Structures: Queue

کیو FIFO پر چلتی ہے: جو پہلے آیا وہ پہلے نکلے۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 18 of 24

★ Golden

Linear Data Structures: Queue ★ Golden

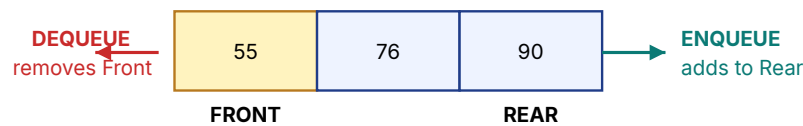
LEARN IT

A **queue** organises data on the **FIFO** principle: **First-In, First-Out**. Whoever arrives first is served first. Order truly matters.

- **ENQUEUE** — add an element at the **Rear** (back).
- **DEQUEUE** — remove an element from the **Front**.

Real-life example: several documents sent to a printer. The printer prints the first document received, then the next.

Queue (FIFO)



Next ENQUEUE joins after 90. Next DEQUEUE removes 55.

Three jobs waiting. New jobs join at the rear; service always takes the front.

WORKED EXAMPLE

Worked example: ENQUEUE / DEQUEUE trace. Start empty. ENQUEUE 55, 76, 90, then ENQUEUE D, then DEQUEUE.

Step	Operation	Queue (Front → Rear)	Front	Rear	Removed
0	(empty)	[]	—	—	
1	ENQUEUE 55	[55]	55	55	
2	ENQUEUE 76	[55, 76]	55	76	
3	ENQUEUE 90	[55, 76, 90]	55	90	
4	ENQUEUE D	[55, 76, 90, D]	55	D	
5	DEQUEUE	[76, 90, D]	76	D	55

```
from collections import deque
q = deque()
q.append(55)
q.append(76)
q.append(90)
q.append("D")
print("after ENQUEUE D:", list(q))
print("DEQUEUE:", q.popleft())
print("left:", list(q))
```

OUTPUT

```
after ENQUEUE D: [55, 76, 90, 'D']
DEQUEUE: 55
left: [76, 90, 'D']
```

append is ENQUEUE at the rear; popleft is DEQUEUE from the front. The first DEQUEUE is 55 (the oldest), never D.

REMEMBER IT

FIFO = First In First Out = a canteen line. Front = the person being served. Rear = the person who just joined.

COMMON MISTAKE

Mistake: DEQUEUEing from the rear, or ENQUEUEing at the front.

Fix: new items only at Rear; removal only at Front. If your first removal is D, you treated the queue as a stack.

اردو میں سمجھیں

کیو FIFO پر چلتی ہے: جو پہلے آیا وہ پہلے نکلے۔ پیچھے (Rear) شامل کرنا ENQUEUE، آگے (Front) سے ہٹانا DEQUEUE ہے۔ پرنٹر کی قطار: جو فائل پہلے بھیجی گئی وہ پہلے چھپے گی۔ پہلی DEQUEUE ہمیشہ سب سے پرانا عنصر نکالتی ہے۔

EXAM TIP

Define queue + FIFO, define ENQUEUE (rear) and DEQUEUE (front), give the printer example, draw the labelled Front/Rear diagram, and trace a short sequence. Examiners mix stack and queue on the same paper: if they say "first come, first served" write Queue; if they say "Undo" write Stack.

CHECK YOURSELF

1. Expand FIFO. Where is an item added, and where is it removed?
2. After ENQUEUE 55, 76, 90, what does DEQUEUE return?
3. Why is a printer a queue, not a stack?

Answers: 1. First-In, First-Out; added at Rear, removed from Front. 2. 55. 3. Documents must print in the order they arrived (first received, first printed), not last-received first.

Next lecture: 2.5.2 Non-Linear Data Structures: Tree